Vignesh Pathak

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Education

University of Illinois Chicago *Aug 2025 – May 2027*
M.S. in Computer Science, GPA: 3.71
Coursework: Computer Algorithms, Intro to Data Science, Computability and Complexity, Computational Biology, Natural Language Processing, Vector Databases

G.H. Raisoni College of Engineering, Nagpur, India *June 2018 – May 2022*
B.E. in Computer Science, CGPA: 8.63

Professional Experience

Southern Illinois Healthcare *Sept 2026 – Present*
AI Full Stack Engineering Co-Op

- Built a production RAG system for organizational knowledge search across a distributed multi-service backend (Django, Go, FastAPI), implementing 14 features including hybrid search and query caching.
- Fixed Excel file search end-to-end, tracing the issue across query handling, relevance boosting, and source classification to its root cause.
- Authored the vendor access and integration checklist for migrating the platform's search index, storage, and OAuth integrations to a new health system.
- Engineered custom Azure document-extraction and text-chunking pipelines to index full multi-page PDFs (30K+ characters), enabling accurate citation-backed answers.
- Use Claude Code daily as an AI pair programmer across the 1-Search codebase to scaffold features, debug production incidents, and review changes, meaningfully speeding up delivery of new search capabilities.

UI Health, Chicago, IL *Nov 2025 – Aug 2026*
AI Engineer & Software Developer, Graduate Research Assistant

- Developed the 1-Search platform across a distributed set of services (Django REST, React, Go, Elasticsearch), transforming the ingestion pipeline from 30-minute polling to an event-driven architecture.
- Built an enterprise cloud data connector with async webhook handling and mutex-based concurrency control, achieving 99.9% reliability and sub-10-second sync latency across distributed components.
- Diagnosed and fixed a production real-time-sync outage caused by Google Drive webhook expiry, shipping a self-renewing webhook cycle and a manual re-registration endpoint.
- Implemented Redis caching enabling 30x faster duplicate searches (under 0.1s) through optimized key strategies.

Accenture, Pune, India *Nov 2022 – Jul 2025*
Data & AI Engineering Analyst

- Built RAG-based semantic search systems using PyTorch and vector embeddings for Fortune 500 retail clients (Macy's, Home Depot, Walmart, Costco), replacing slower keyword-based search.
- Designed and deployed agentic AI workflows using CrewAI, automating multi-step retail data processing and replacing manual workflows with autonomous agent pipelines.
- Developed and fine-tuned deep learning models in PyTorch for retail forecasting and optimization, improving prediction quality across client engagements.
- Implemented vector search infrastructure supporting large-scale semantic search across high-volume retail product catalogs.
- Delivered end-to-end AI search and retrieval solutions across multiple retail engagements, combining RAG, vector search, and agentic orchestration.

Capgemini, Mumbai, India *Jan 2022 – Apr 2022*
Java Full Stack Developer Intern

- Developed and maintained web application features using Angular on the frontend and Spring Boot on the backend during a Java full-stack engineering internship.
- Built and integrated RESTful APIs using Spring Boot, connecting backend services with Angular-based frontend components.
- Completed structured training in Java, Spring Boot, and Angular, gaining hands-on full-stack development experience.

Projects

VelocityLLM *Self-hosted API gateway – Go, Docker, Kubernetes, distributed systems*

- Designed and built a self-hosted API gateway in Go with a pluggable routing strategy engine, per-service circuit breakers, and cost-tiered fallback chains for production reliability.
- Implemented a full OAuth2 authorization-code flow (Google + GitHub) with CSRF-protected state validation, JWT-based auth with role-based middleware, and SHA-256-hashed API key management with rotation and per-key rate limits.
- Authored production HAProxy and Nginx configs and Kubernetes manifests with readiness/liveness probes, deploying via a multi-stage Docker build for container orchestration.
- Used Claude Code daily across the project to implement the routing engine, debug Go toolchain quirks, and iterate on the MCP server, significantly compressing solo development time.

Noto *Autonomous code-review agent – Python, FastAPI, CI/CD, distributed async*

- Built a FastAPI backend with 20 REST/SSE endpoints across 7 routers, backed by PostgreSQL with 8 schema migrations managed through an async ORM.
- Built async job processing with Redis-backed workers decoupling webhook ingestion from LLM-heavy review work, plus a pub/sub-backed SSE endpoint streaming live progress to a dashboard.
- Set up a GitHub Actions CI pipeline enforcing ruff linting, strict mypy type-checking, and a 70% pytest coverage gate against real Postgres and Redis containers, deployed via Docker.
- Used Claude Code daily as a standing pair programmer to design the multi-agent pipeline, write the 149-test suite, and triage CI failures throughout development.